AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW RESOURCE LOCAL LANGUAGES

R26-DS-011

Project Proposal Report

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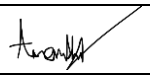
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DECLARATION

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ABSTRACT (Maximum 300 words)

Recurrent floods, landslides and multi-hazard disasters in Sri Lanka created datasets of hundreds of emergency calls and SMS messages within minutes. However, the Disaster Management Centre (DMC) does not yet have an automated system that can intelligently analyze these reports in multiple languages, recognize which reports refer to the same incident and provide merged situational intelligence to decision-makers on-line. During the floods in Western Province of 2016 and 2017 emergency operators were inundated with duplicate and conflicting reports causing delays in resources deployment that led to preventable deaths. This work fills that important operational void."

Specific research gap is the multi-source multilingual incident fusion for low-resource South Asian languages. Current emergency deduplication systems work only on English data, do not vary a threshold similarity depending on the urgency of an incident and are unable to process Sinhala or Tamil variants of the same location name developed in a Roman script. No previous work had used cross-lingual sentence embeddings for emergency report clustering in Sri Lanka, nor have existing Emergency Response Management systems employed urgency-sensitive merging thresholds as a component of architectural safety.

In this work, we present Component 4 of a four-component AI-based emergency triage system: the Multi-Source Incident Fusion (MSIF), Unified Incident Records (UIR), and Real-Time Operator Dashboard module. It consists of voice call and SMS triage pipeline ingests structured JSON (Components 1 and 3), computing cross-lingual semantic similarity using language-agnostic BERT embeddings (LaBSE), it then clusters input reports according to Urgent epsilon parameters in DBSCAN. For critical incidents an increased threshold of evidence is needed to merge, establishing an architectural safety asymmetry. A multilingual location gazetteer for Sri Lanka resolves place names in every possible orthography to gps coordinates. We implement conflict detection with weighted resolution and a real-time WebSocket-driven dashboard.

Technologies that are used: LaBSE (sentence-transformers), DBSCAN (scikit-learn), FastAPI, PostgreSQL with PostGIS, Redis, React. js, Leaflet. js, and OpenStreetMap. The anticipated results are a clustering accuracy rate greater than 90%, a recall rate greater than 85%, an Critical incident false-merge rate less than 2%, the latency of the end-to-end pipeline less than 8 s and operator action percentage below to 15%. From a social standpoint, the system helps reduce response time during mass-casualty situations and enhances disaster resilience architecture in Sri Lanka. The system is commercially leveraging pilot deployment with the DMC and scalable licensing to other South Asian national disaster management authorities.

Keywords: Multi-Source Incident Fusion, Urgency-Sensitive Clustering, LaBSE, DBSCAN, Unified Incident Records, Real-Time Dashboard, Disaster Management, Sinhala, Tamil, Sri Lanka.

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TABLE OF CONTENT (Auto Generated)

DECLARATION 2

ABSTRACT (Maximum 300 words) 3

TABLE OF CONTENT (Auto Generated) 5

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

1. INTRODUCTION 13

1.1 Background and Context 13

1.2 Literature Review 14

1.3 Research Gap 15

2. OBJECTIVE 17

2.1 Main Objective 17

2.2 Specific Objectives 17

3. METHODOLOGY 18

4. HIGH-LEVEL SYSTEM ARCHITECTURE 20

4.1 Overview of proposed solution and its components 20

4.2 System diagram of components 21

4.3 Highlight the individual component and its parts 21

4.4 Component interactions 22

4.5 Data Flow Explanation 23

4.6 Scalability and Security Considerations 23

5. USER REQUIREMENTS 24

5.1 Stakeholder Identification 24

5.2 Functional Requirements 24

5.3 Non-Functional Requirements 25

6. COMMERCIALIZATION PLAN 26

8. WORK BREAKDOWN STRUCTURE (WBS) 28

9. GANTT CHART 31

10. REFERENCES LIST 32

Appendix A: Risk Analysis 34

Appendix B: UI Mockup 35

LIST OF TABLES

Table 1:List of Abbreviations..... 12
Table 2:Overview proposed Solution's Components 20
Table 3 :Component 4 Layers 22
Table 4:Budget and Justification..... 27
Table 5:Work Breakdown Structure 30
Table 6 :Risk analysis..... 34

LIST OF FIGURES

Figure 1:High Level System Architecture 21
Figure 2: Gantt Chart..... 31
Figure 3:UI Mockup..... 35

LIST OF ABBREVIATIONS

Abbreviation	Full Term	Where Used / Notes
AI / Machine Learning		
AI	Artificial Intelligence	General term; used throughout all chapters
ML	Machine Learning	Subset of AI; referenced in methodology
NLP	Natural Language Processing	Core technology across all four components
NER	Named Entity Recognition	Used in C1 and C3 to extract location, incident type, people count
ASR	Automatic Speech Recognition	Core sub-module of Component 1 (voice pipeline)
BERT	Bidirectional Encoder Representations from Transformers	Foundation model architecture; parent of LaBSE, mBERT
mBERT	Multilingual BERT	Compared against LaBSE in literature review; baseline model
LaBSE	Language-Agnostic BERT Sentence Embeddings	Core cross-lingual embedding model used in C4 (Feng et al., 2022)
TF-IDF	Term Frequency – Inverse Document Frequency	Monolingual baseline method compared against LaBSE in evaluation
LLM	Large Language Model	Used in Component 2 AI emergency agent (GPT-4o)
SUS	System Usability Scale	Standard usability evaluation instrument; used in DMC operator drill
F1	F1 Score (Harmonic Mean of Precision and Recall)	Primary clustering evaluation metric
Algorithms & Methods		
DBSCAN	Density-Based Spatial Clustering of Applications with Noise	Clustering algorithm used in C4 Stage 5 (Ester et al., 1996)
DSR	Design Science Research	Research methodology adopted for this project (Hevner et al., 2004)

GPS	Global Positioning System	Coordinate system used for location normalization in the gazetteer
ISO	International Organization for Standardization	ISO 8601 UTC timestamp format used in canonical schema
UTC	Coordinated Universal Time	All timestamps in C4 normalized to UTC
L2	L2 Norm (Euclidean Norm)	LaBSE embeddings are L2-normalized; enables cosine similarity = dot product
System Components (This Project)		
C1	Component 1 — Voice Pipeline	Voice language ID, ASR, urgency detection, and incident NER
C2	Component 2 — AI Emergency Agent	LLM-based conversational gap-filling for incomplete C1 records
C3	Component 3 — SMS Intelligence Pipeline	Multilingual SMS normalization, NER, and urgency classification
C4	Component 4 — Multi-Source Incident Fusion	This research component — fusion, UIRs, and operator dashboard
UIR	Unified Incident Record	Canonical deduplicated record for each real-world incident in C4
EOC	Emergency Operations Centre	Physical DMC control room where operators use the C4 dashboard
Software, Infrastructure & Protocols		
API	Application Programming Interface	Used for all inter-component communication (REST + WebSocket)
REST	Representational State Transfer	HTTP API style used for operator action endpoints
HTTP	Hypertext Transfer Protocol	Transport protocol for C1/C2/C3 → C4 webhook POST requests
HTTPS	Hypertext Transfer Protocol Secure	TLS-encrypted HTTP; required for all C4 external communication
WSS	WebSocket Secure	TLS-encrypted WebSocket; used for C4 → dashboard real-time broadcast

TLS	Transport Layer Security	Encryption protocol; TLS 1.3 enforced for all C4 connections
JWT	JSON Web Token	Authentication token issued on operator login; 1-hour expiry
JSON	JavaScript Object Notation	Data format for all inter-component reports and UIR serialization
SQL	Structured Query Language	Query language for PostgreSQL UIR database
GIS	Geographic Information System	Spatial data framework; PostGIS extends PostgreSQL with GIS support
AOF	Append-Only File	Redis persistence mode used to protect the report ingestion queue
WAL	Write-Ahead Log	PostgreSQL crash recovery mechanism protecting committed UIR records
FIFO	First In, First Out	Redis queue processing order for incoming reports
CSRF	Cross-Site Request Forgery	Web security threat; CSRF protection applied to C4 REST action endpoints
DoS	Denial of Service	Security threat; mitigated by rate limiting and Redis queue buffering
LDAP	Lightweight Directory Access Protocol	DMC staff directory used for operator dashboard authentication
JWT	JSON Web Token	Issued on LDAP login; required for all operator API calls
gRPC	Google Remote Procedure Call	High-performance protocol used between C4 workers and GPU LaBSE server
GPU	Graphics Processing Unit	Used for accelerated LaBSE inference under high disaster load
CPU	Central Processing Unit	Default LaBSE inference hardware during low-load peacetime operation
AES	Advanced Encryption Standard	AES-256 used for PostgreSQL database encryption at rest

OOM	Out of Memory	Failure mode monitored by health check; triggers LaBSE server alert
Organisations & Domain Context		
DMC	Disaster Management Centre	Primary customer and deployment target; operates the 117 hotline
SLIIT	Sri Lanka Institute of Information Technology	Sponsoring academic institution
NSF	National Science Foundation (Sri Lanka)	Primary proposed funding source (LKR 1.5M)
NDMA	National Disaster Management Authority	Bangladesh equivalent of DMC; identified as regional export market
NDRF	National Disaster Response Force	India equivalent; identified as regional export market
NIPO	National Intellectual Property Office of Sri Lanka	Patent filing body for urgency-sensitive DBSCAN algorithm IP
SDG	Sustainable Development Goal	UN SDG 11 (Sustainable Cities) and SDG 3 (Good Health) alignment
UNDRR	United Nations Office for Disaster Risk Reduction	Sendai Framework body; referenced in background and SDG alignment
EOC	Emergency Operations Centre	Physical facility housing DMC operators using the C4 dashboard
GovTech	Government Technology	Commercialization category for the proposed SaaS licensing model
SaaS	Software as a Service	Revenue model proposed for DMC national deployment
IP	Intellectual Property	Refers to the three IP assets: gazetteer, algorithm patent, paper
LKR	Sri Lankan Rupee	Currency used in all budget figures
USD	United States Dollar	Currency used in regional export market pricing
Evaluation & Performance Metrics		

FMR	False-Merge Rate	Key safety metric: % of distinct Critical incidents incorrectly merged (target < 2%)
P	Precision	Clustering metric: % of merged report pairs that are truly the same incident (target > 90%)
R	Recall	Clustering metric: % of same-incident pairs correctly merged (target > 85%)
PII	Personally Identifiable Information	Caller phone numbers, names — excluded from UIR storage by design
QoS	Quality of Service	System performance characteristic; referenced in non-functional requirements
SLA	Service Level Agreement	Implied by latency and uptime targets in non-functional requirements (< 8 sec, 99.5% uptime)

Table 1:List of Abbreviations

1. INTRODUCTION

1.1 Background and Context

Sri Lanka is one of the top disaster-hit countries in South Asia. Biogeochemical context: Recurrent floods in the country (average 8–12 significant flood events per year) [1], landslides, cyclones and compound multi-hazard events due to southwest and northeast monsoons add to this uncertainty. The May 2017 Southwest Monsoon floods and landslides caused over 200 deaths, the disappearance of around 80 persons and the displacement of 600,000 people in 15 districts [2]. Emergency calls to the Disaster Management Centre (DMC) reached several hundred an hour in the first 24 hours, in three languages — Sinhala, Tamil and English — but from callers using dialect forms of each language, Romanized script and then landmark references to their locations and situations.

Disaster Management Center (DMC) which was formed by the Disaster Management Act No. 13 of 2005, has the task of national disaster response simulator. Its Emergency Operations Centre (EOC) fielded calls on the 117 hotline, but there was a manpower bottleneck: call-takers manually logged incidents and automated deduplication was propped up at best. If 15 callers within eight minutes report the same flood at Kelaniya — each using different language or location references — the EOC generates 15 distinct records. Operators waste valuable response time calling out duplicates instead of getting resources on the road. This information glut is an established operational failure state during Sri Lankan disaster response [3].

Victims of a disaster waiting for an emergency response are the most affected. Other stakeholders are DMC operators who need to triage hundreds of reports in very short time. The wider Sri Lankan public suffers through systematic under-resourcing of response — when operators become overwhelmed, allocation is delayed and mismanaged. Sri Lanka is home to 22 million people, some half of whom live in flat coastal deltas prone to flooding [4]. The average economic loss caused by annual flood events throughout the country can be estimated at LKR 50–120 billion per year, and casualties usually number in the hundreds per annum [5].

This is a global concern. The United Nations Office for Disaster Risk Reduction (UNDRR) referring to the Sendai Framework for Disaster Risk Reduction 2015–2030 emphasizes on early warning and disaster risk information as one of the Target G investments. Sri Lanka as a nation will have to build-up systematically towards achieving these requirements and enhancing the capacity to process such information at the most optimal level as facilitated through this system.

The Computer-Aided Dispatch (CAD) system used in high income countries such as the Commercial off-the-shelf (COTS) products from Motorola PremierOne, Hexagon I/CAD, Tyler Technologies New World is not suitable. These systems document the structured information of each incident and assume a single language (usually English), a pre-geocoded database of addresses, and do not have the capability to identify that multiple calls pertain to a single event. These systems are also based on assumptions of the stable infrastructure of urban areas and do not account for the multilingual, unstructured nature of disaster reporting common in South Asia [6].

While designed for social media-based disaster intelligence systems like CrisisNLP, AIDR, and Tweedr that monitor Twitter and Facebook to extract information about disaster events from users' posts, our focus is different. These systems only work with English language and public social media, and exclude emergency telephone calls and private SMS messages sent by affected people to authorities. Moreover, they only provide unstructured event discovery and do not output structured data required for operational-level dispatch decisions [7]. Other research systems have proposed different solutions to extract disaster-related information. For instance, Belcastro et al. in [8] employ DBSCAN clustering algorithm on the geotagged social media posts for sub-event detection in the aftermath of a disaster. However, these systems depend on only the location coordinates of the social media posts and the semantic deduplication is restricted to English language. The CrisisTransformers family of models introduced by Lamsal et al. in [9] provide the state of the art models in terms of crisis sentence encoders but are only for the English language. In all, the two scripts spoken by a

majority of the population in Sri Lanka i.e., Sinhala and Tamil are completely lacking in the published emergency NLP literature.

This research presents Component 4 of a four-part emergency triage system (R26-DS-011). The four parts are: C2 (voice language identification, ASR, and incident extraction), C2 (multilingual AI emergency agent), C3 (SMS/text emergency intelligence), and C4 (multi-source incident fusion, UIRs, and real-time dashboard). Component 4 takes structured JSON data from C2 and C3, merges reports into deduplicated Unified Incident Records (UIRs) using LaBSE cross-lingual embeddings and urgency-sensitive DBSCAN clustering. It then provides live situational intelligence to DMC operators through a real-time WebSocket dashboard.

This research directly supports United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities, Target 11.5: reduce deaths and economic losses from disasters) and SDG 3 (Good Health and Well-Being, specifically reducing preventable deaths from delayed emergency response). It aligns with the SLIIT Research Cluster on Artificial Intelligence and Data Science, specifically the sub-cluster on AI for Social Good and disaster resilience systems.

1.2 Literature Review

The literature associated with Component 4 includes the following domains: emergency event clustering from crowdsourced data, multilingual crisis NLP, deduplication methods, and real-time disaster management systems.

In the work done by Belcastro et al. [8], the authors proposed a method called SEDOM-DD, which is based on the DBSCAN algorithm to identify the occurrence of a disaster's sub-events from geotagged social media data. The proposed system uses spatial clustering with a fixed epsilon to identify the geographic concentration of related social media posts. However, the proposed method only relies on the geographic coordinates of the posts to compute the clusters, without any semantic similarity calculation. In addition, the proposed method relies on a fixed epsilon value to merge the clusters, which does not consider the criticality of the event to be clustered.

A framework for emergency event detection and clustering of Chinese Sina Weibo posts was proposed by Huang et al. [10]. The framework, BERT-Att-BiLSTM, was able to detect 3W attributes (what, where, when) and perform dynamic clustering based on text similarity, with an accuracy of 94.15%. The framework is strictly monolingual and mono-source, meaning it cannot handle code-switching in multilingual texts or combine voice and SMS data streams. There was no urgency-sensitive clustering.

In a recent study, Lamsal et al. proposed a model called CrisisTransformers, where the model is a collection of pre-trained transformer-based models trained over a massive corpus of 15 billion tokens from over 30 crisis events, performing better than the general SBERT model by a margin of 17.43% over 18 crisis-specific datasets. The authors have clearly stated that their sentence encoder "remains monolingual, processing only English texts."

In another study by the same researchers, the gap is addressed by proposing XLM-R and mBERT-based sentence encoders, but the languages of the crisis in Europe, i.e., Sinhala and Tamil, are still not addressed.

Sanchez et al. [12] have already explored the problem of cross-lingual crisis classification with mBERT and knowledge distillation on 53 crises and 160,000+ messages. The paper proves the possibility of cross-lingual transfer for crisis NLP, but the classification is on a single message basis rather than incident formation based on multiple reports. There are already papers (Wu et al. [13], Are All Languages Created Equal [14]) proving the inferiority of mBERT on under-resourced languages with small Wikipedia sizes; Sinhala has less than 12,000 articles on Wikipedia.

Feng et al. [15] have recently proposed a language-agnostic BERT sentence embedding model called LaBSE, which is pre-trained on 109 languages using a dual encoder model and a translation ranking objective function. LaBSE supports Sinhala (si) and Tamil (ta) languages and generates L2-normalized 768-dimensional vectors in a shared space where semantically equivalent sentences across any language have similar vectors. This makes it the only pre-trained model applicable to the Sinhala-Tamil-English cross-lingual clustering task in C4 without fine-tuning.

Mukhopadhyay et al. [6] surveyed 50 years of Emergency Response Management research, including incident prediction, detection, resource allocation, and CAD dispatch. The research gaps on multi-source incident fusion and multilingual processing, which are not addressed by previous ERM research, are directly addressed by this research.

Agouris et al. [16] studied the role of GIS-based disaster management information systems, highlighting the need to normalize locations based on diverse naming conventions. This has implications for the Sri Lanka emergency location gazetteer in C4, which has to cope with Sinhala Unicode, Tamil Unicode, Romanization, and landmark forms to GPS coordinates.

The International CAD Systems literature [17] reports that current dispatch platforms treat each call as a separate entity, and there is no deduplication across calls. The dispatcher workload peaks of 200 to 400 calls per hour during a major incident are a known system failure. Automated deduplication is a known research and engineering problem.

Research into the Sri Lankan model of responding to disasters [2][3][5] has consistently concluded that the major cause of preventable casualties is the delay in the response effort and the misallocation of resources, with information overload in the first 2-4 hours of a disaster being identified as the major weakness. There is no technical solution to specifically address this issue.

1.3 Research Gap

The following five gaps, evidenced by the literature reviewed above, collectively justify this research:

Gap 1: No multilingual emergency incident fusion system exists for Sinhala, Tamil, or any South Asian low-resource language. All published emergency clustering systems are English-only or operate on high-resource languages. LaBSE's Sinhala/Tamil support has never been evaluated for emergency incident deduplication.

Gap 2: All existing emergency clustering systems apply a single fixed similarity threshold regardless of incident urgency. The cost asymmetry between a false merge on a Critical incident (potentially fatal) versus a false split (one extra operator click) is never formalized as an architectural constraint.

Gap 3: No system fuses structured reports from multiple input channels (voice ASR pipeline + SMS NLP pipeline) into deduplicated Unified Incident Records. Existing systems are single-source.

Gap 4: No Sri Lankan emergency location resource exists. No published gazetteer maps Sinhala Unicode, Tamil Unicode, Romanized Sinhala, Romanized Tamil, and English forms of Sri Lankan place names to GPS coordinates.

Gap 5: Human-in-the-loop conflict resolution with a full operator authority model (confirm, split, merge, correct, dismiss) for automatically-fused emergency records has not been implemented or evaluated in any published system.

These gaps are not merely incremental improvements over existing work — they represent the absence of any published solution applicable to the Sri Lankan emergency management context. The proposed system is the first to address all five gaps simultaneously.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a multilingual multi-source incident fusion system for Sri Lankan disaster emergency reports that correctly deduplicates Sinhala, Tamil, and English reports into unified incident records using urgency-sensitive LaBSE-based DBSCAN clustering, achieving clustering precision above 90%, recall above 85%, and a Critical incident false-merge rate below 2%, and delivers fused situational intelligence to operators through a real-time dashboard with full pipeline latency under 8 seconds.

2.2 Specific Objectives

1. To construct a multilingual Sri Lanka emergency location gazetteer mapping 5,000+ place name variants across Sinhala Unicode, Tamil Unicode, Romanized Sinhala, Romanized Tamil, English, and landmark references to GPS coordinates, achieving location normalization coverage above 85% on the evaluation corpus.
2. To implement a combined three-dimensional similarity scoring function (0.5 × semantic + 0.3 × geographic + 0.2 × temporal) using LaBSE embeddings and Haversine distance, demonstrating that LaBSE achieves at least 15% higher clustering F1 than a TF-IDF monolingual baseline on cross-language incident report pairs.
3. To develop and empirically tune an urgency-sensitive DBSCAN clustering algorithm with differential epsilon values (CRITICAL: 0.15, HIGH: 0.25, MEDIUM: 0.30, LOW: 0.40) via grid search, maintaining Critical incident false-merge rate below 2% while maximizing overall clustering F1.
4. To implement automated conflict detection and three-factor weighted resolution (source reliability, corroboration, recency) for contradictory fields in merged incident reports, achieving conflict detection accuracy above 80% on annotated evaluation scenarios.
5. To build and deploy a real-time WebSocket operator dashboard (React.js + Leaflet.js/OpenStreetMap) that displays all active UIRs with urgency-coded map pins, full incident timelines, conflict flags, and a five-action operator interface, with full pipeline latency under 8 seconds and UIR update latency under 5 seconds.
6. To evaluate the complete system through 20 scripted multi-caller, multi-language disaster scenarios and a live DMC operator drill, achieving operator action rate below 15% and collecting System Usability Scale (SUS) feedback from DMC operators.

3. METHODOLOGY

This research adopts Design Science Research (DSR) methodology (Hevner et al., 2004), which is appropriate for research that produces a novel IT artifact as its primary contribution. DSR proceeds through three cycles: the relevance cycle (establishing the problem from the DMC operational context), the design cycle (iterative artifact construction and refinement), and the rigor cycle (grounding design decisions in the literature). The research artifact is the Component 4 software system. Evaluation is conducted through both controlled laboratory testing (20 scripted scenarios with ground-truth labels) and a field drill with real DMC operators.

All incoming JSON reports from Components 1 and 3 are converted to a canonical schema by the normalize() function before any fusion logic runs. This function handles field aliasing (location_raw vs. loc_text), type coercion (urgency string/integer normalization), value clamping (confidence [0,1], people_involved [0,9999]), timestamp parsing with ISO 8601 fallback to receive_time, and flag assignment (low_confidence, no_location, no_timestamp). The function is idempotent and stateless.

A curated JSON database maps every known Sri Lankan place name variant to a canonical name and GPS coordinate. A three-stage lookup algorithm is applied: (A) exact match via a pre-built O(1) Python dictionary; (B) fuzzy match using Levenshtein distance via rapidfuzz with a configurable similarity threshold (default: 80/100); (C) landmark pattern extraction via regex followed by recursive Stage A/B lookup. Unresolvable locations are flagged rather than dropped, with geographic_score forced to 0.0 for downstream clustering.

The LaBSE model (Feng et al., 2022 [15]) generates 768-dimensional L2-normalized vectors from concatenated incident text (incident_type + location_raw + key_phrases). LaBSE supports Sinhala, Tamil, and English in a shared embedding space, enabling cosine similarity to measure semantic equivalence across languages with no fine-tuning. The model is loaded once at startup from the sentence-transformers library.

A three-dimensional combined similarity score is computed: S(r,u) = 0.5 * S_sem + 0.3 * S_geo + 0.2 * S_time, where S_sem is LaBSE cosine similarity, S_geo is 1 - min(haversine_km/5.0, 1.0), and S_time is exp(-0.05 * delta_t_minutes). An incremental variant of DBSCAN assigns incoming reports to existing UIR clusters (or creates new UIRs) using urgency-adjusted epsilon values: CRITICAL=0.15, HIGH=0.25, MEDIUM=0.30, LOW=0.40. Critical reports additionally require a five-condition safety gate before merging (semantic >0.88, distance <0.8 km, time gap <15 min, corroborating sources >=2, exact incident type match).

When a new report merges into an existing UIR, contradiction detection compares numeric fields (>30% deviation triggers a conflict) and categorical fields (exact mismatch). Resolution applies three weighted factors in priority order: source reliability (voice > SMS), corroboration (>= 2 independent sources agreeing), and recency. Unresolvable conflicts produce a display range (e.g., 3-10 people) and set the operator_review_required flag.

Each cluster is represented as a Unified Incident Record stored in PostgreSQL with PostGIS. UIRs are updated atomically on each new merge. A WebSocket server (FastAPI) broadcasts updated UIRs to all connected React.js dashboard clients within 1.5 seconds of the UIR update completing. The dashboard renders urgency-coded pins on a Leaflet.js/OpenStreetMap map, a sortable incident list (Critical incidents pinned top), full UIR detail panels, and a five-action operator interface (Confirm/Split/Merge/Correct/Dismiss) with full audit logging.

The C4 component is implemented using a set of carefully selected tools and platforms that support real-time processing, multilingual understanding, and scalable incident management. LaBSE (sentence-transformers) is used for cross-lingual embeddings because it is one of the few pre-trained models that can represent Sinhala, Tamil, and English in a shared semantic vector space without requiring additional fine-tuning, enabling accurate similarity comparison between reports written in different languages. For incident grouping, DBSCAN from the scikit-learn library is used because it does not require a predefined number of clusters and naturally treats unrelated reports as noise, while its adjustable epsilon parameter allows the clustering radius to be tuned based on the urgency of incidents. Rapidfuzz is employed for fuzzy location matching since it provides one of the fastest Python implementations of the Levenshtein distance algorithm, and its WRatio scorer effectively handles spelling variations, partial

matches, and common Romanized Sinhala place names. The backend processing layer is implemented using FastAPI, which provides native asynchronous support, built-in WebSocket functionality, and extremely fast request handling, making it well suited for real-time webhook ingestion from upstream components and live updates to the dashboard. For persistent storage, PostgreSQL combined with the PostGIS extension is used to store Unified Incident Records (UIRs) while enabling efficient geographic queries such as distance calculations required for spatial clustering. Redis acts as an in-memory report ingestion queue that absorbs sudden spikes in incoming reports during disaster situations, ensuring that no data is lost even if processing temporarily lags behind the arrival rate. On the frontend, the operator dashboard is developed using React.js 18, which provides a component-based architecture and efficient reactive rendering for updating only the parts of the interface affected by incoming WebSocket events. Finally, Leaflet.js integrated with OpenStreetMap is used to visualize incidents on a map, as it offers free access to detailed Sri Lankan map data, avoids the API costs associated with Google Maps, and supports custom markers that can display incidents with different colors based on urgency levels.

The evaluation dataset consists of 20 scripted disaster scenarios totalling approximately 300 simulated reports, all synthetically generated by the researcher. No real victim data, personal information, or private DMC call records are used. All voice and SMS content in the evaluation scenarios is fictional and constructed to represent realistic disaster report patterns based on published descriptions of past Sri Lankan flood events and field linguist consultation.

The DMC operator drill in Month 9-10 involves professional DMC staff interacting with the dashboard using the synthetic evaluation scenarios only. Participant data is anonymized and used solely for system usability evaluation. Informed consent will be obtained from all participants. Ethical approval will be obtained from the SLIIT Research Ethics Committee prior to any human participant involvement. No biometric data, location tracking, or personal identifiers are collected.

The validation strategy for the C4 component focuses on evaluating clustering accuracy, system reliability, and real-time operational performance using clearly defined metrics and target thresholds. Clustering precision measures the percentage of merged report pairs that actually correspond to the same real-world incident, with a target of greater than 90%, and is evaluated by comparing system-generated clusters with manually created ground-truth labels. Clustering recall evaluates the percentage of true same-incident report pairs that the system successfully merges, with a target of above 85%, also determined through comparison with the labelled ground-truth dataset. To ensure safety in critical situations, the critical false-merge rate measures how often distinct Critical incidents are mistakenly merged into the same cluster; this must remain below 2% and is tested using specially designed adjacent-incident scenarios. Location normalization coverage assesses the percentage of location strings that can be successfully resolved into GPS coordinates, targeting more than 85%, which is measured by running the location gazetteer on the evaluation corpus. System responsiveness is measured through full pipeline latency, defined as the total time from when a JSON report arrives to when the processed result appears on the dashboard, with a target of less than 8 seconds, calculated by time stamping each stage of the pipeline. The system's ability to identify inconsistent reports is evaluated using conflict detection accuracy, which measures the percentage of annotated contradictions correctly flagged by the system, targeting over 80% using annotated evaluation scenarios. Operational efficiency is further assessed through the operator action rate, representing the percentage of Unified Incident Records (UIRs) that require manual correction by operators, with a target of below 15%, measured using Disaster Management Centre (DMC) operator drill logs. Finally, the effectiveness of multilingual semantic understanding is measured through the LaBSE vs. TF-IDF F1 gain, where the system should achieve at least a 15% improvement in F1 score when using LaBSE embeddings compared to a monolingual TF-IDF baseline on the same evaluation dataset.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of proposed solution and its components

The R26-DS-011 system is a four-component AI-based emergency triage pipeline designed to transform raw multilingual disaster calls and SMS messages into fused, operator-ready situational intelligence within eight seconds. Each component is independently deployable but produces structured output consumed by the next. The full pipeline spans from caller speech to a live operator dashboard.

Component 4 (this research contribution) is the final intelligence integration layer. It sits between the upstream NLP pipelines (C2 voice, C3 SMS) and the human operators at the Disaster Management Centre. It does not process raw audio or text — it receives structured JSON, fuses it, and presents unified incident intelligence in real time.

Component	Name	Input	Output	Feeds Into
C1	Voice Language ID, ASR & Incident Extraction	Audio from 117 hotline calls	Completed incident record; fills missing fields via AI dialogue	**C2**
C2	Multilingual AI Emergency Agent	C1 partial output + caller audio	Completed incident record; fills missing fields via AI dialogue	**C4**
C3	Low-Bandwidth SMS/Text Emergency Intelligence	SMS messages to emergency number	Structured JSON: incident_type, location_raw, urgency, key_phrases, confidence	**C4**
C4 ★	**Multi-Source Incident Fusion, UIRs & Real-Time Dashboard**	Structured JSON from C2, C3	Deduplicated UIRs → Live operator dashboard	**DMC Operators**

Table 2:Overview proposed Solution's Components

4.2 System diagram of components



Figure 1:High Level System Architecture

4.3 Highlight the individual component and its parts

Multi-Source Urgency Base Incident Fusion, UIRs & Real-Time Dashboard

Component 4 is the central intelligence hub of the system. It receives structured JSON from C2, and C3, and performs all fusion, deduplication, conflict resolution, and real-time dashboard delivery. Its eight internal stages are described fully in Chapter 3. The internal architecture consists of five logical layers:

Layer	Modules	Responsibility
Ingestion Layer	FastAPI Webhook, Redis Queue	Receives JSON from C2/C3. Assigns source_id. Pushes to async processing queue. Handles

		burst traffic during disaster peaks without data loss.
Processing Layer	Normalizer, Gazetteer, LaBSE Encoder, DBSCAN Clusterer	Converts raw reports to canonical schema. Resolves locations to GPS. Generates cross-lingual embeddings. Applies urgency-sensitive DBSCAN to find matching UIR or create new one.
Fusion Layer	Conflict Detector, Weighted Resolver, UIR Manager	Detects contradictions between merged reports. Applies source-reliability/corroboration/recency weighting. Updates UIR fields, confidence score, flags, and chronological timeline.
Storage Layer	PostgreSQL + PostGIS, Redis Cache	Persists all UIRs with ACID guarantees. PostGIS enables geographic proximity queries. Redis caches active UIR embeddings for sub-millisecond similarity lookup. Full operator action audit log stored.
Presentation Layer	WebSocket Server, React.js Dashboard, Leaflet.js Map	Broadcasts UIR updates to all connected operators within 1.5 s. Renders urgency-coded map, sortable incident list, full UIR detail panel, and operator action interface. All actions logged.

Table 3 :Component 4 Layers

4.4 Component interactions

The Component 4 module interacts with multiple subsystems within the overall emergency triage architecture. It receives structured JSON incident reports from Component 2 (Voice-Based Emergency Intelligence) and Component 3 (SMS/Text Emergency Intelligence) through a FastAPI webhook interface. These reports are processed by internal modules including the normalizer, location gazetteer, semantic encoder, and clustering engine to determine whether the report belongs to an existing incident or represents a new event. After incident fusion and conflict resolution, the system updates the corresponding Unified Incident Record (UIR) in the database. The updated incident information is then transmitted to the real-time operator dashboard using WebSocket communication, allowing Disaster Management Centre (DMC) operators to view and manage incidents immediately.

4.5 Data Flow Explanation

The data flow within Component 4 begins when structured emergency reports are received from upstream components as JSON objects. These reports first pass through a normalization stage that standardizes fields such as location, urgency level, and timestamp. Next, the system resolves location information using the Sri Lankan location gazetteer and generates semantic embeddings using the LaBSE model to represent the report content. A similarity scoring function combining semantic, geographic, and temporal features is then calculated to determine whether the report matches an existing incident cluster. If a match is found, the report is merged into the corresponding Unified Incident Record; otherwise, a new incident record is created. The updated incident information is stored in PostgreSQL and instantly broadcast to the dashboard through WebSocket updates.

4.6 Scalability and Security Considerations

The system is designed with scalability and security as key architectural considerations. Scalability is achieved through the use of asynchronous FastAPI services, Redis queues for buffering incoming reports during traffic spikes, and efficient clustering algorithms that can process large volumes of incident reports in real time. The architecture allows horizontal scaling by deploying multiple processing workers when disaster reporting volume increases. Security is maintained by hosting the entire system within a controlled server environment, ensuring that sensitive emergency data is not transmitted to external services. All communications between components use secure API endpoints, and access to the operator dashboard is restricted through authenticated user accounts and role-based permissions to prevent unauthorized access to emergency information.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The C4 component interacts with several key stakeholders, each playing a specific role in the emergency response workflow. DMC EOC Operators are the primary users of the system, as they handle incoming emergency calls, coordinate response teams, and dispatch resources. They interact directly with the real-time dashboard, where they can monitor Unified Incident Records (UIRs) and perform actions such as confirming, splitting, merging, correcting, or dismissing incidents based on verified information. DMC Management consists of senior officials who supervise Emergency Operations Center activities and review operational performance after events. They primarily interact with C4 through post-event audit reports, which are generated from the operator action logs associated with UIR handling. The C2 and C3 components function as upstream pipeline modules that provide structured incident reports to C4. These components send JSON-formatted reports through a FastAPI webhook, enabling C4 to process and cluster the incoming data, although they do not receive direct feedback from C4. Finally, the SLIIT Research Team is responsible for developing, configuring, and evaluating the system. They interact with C4 by setting up evaluation scenarios, analysing system performance metrics such as precision, recall, and latency, and refining the algorithms and system parameters based on experimental results.

5.2 Functional Requirements

1. The system shall receive structured JSON reports from Component 1 (voice) and Component 3 (SMS) via a FastAPI webhook and normalize them to a canonical schema within 0.5 seconds of arrival.

2. The system shall resolve incoming location strings to GPS coordinates by performing exact match, fuzzy match, and landmark pattern lookup against the Sri Lanka emergency location gazetteer, covering Sinhala Unicode, Tamil Unicode, Romanized Sinhala, Romanized Tamil, English, and landmark reference forms.

3. The system shall generate a 768-dimensional LaBSE embedding for each normalized report, supporting Sinhala, Tamil, and English text in a shared vector space.

4. The system shall compute a three-dimensional combined similarity score (semantic 50%, geographic 30%, temporal 20%) between each incoming report and all active UIRs.

5. The system shall assign incoming reports to UIR clusters using urgency-sensitive DBSCAN with epsilon values CRITICAL=0.15, HIGH=0.25, MEDIUM=0.30, LOW=0.40, and shall create a new UIR for reports that match no existing cluster.

6. The system shall apply a five-condition safety gate before merging any Critical urgency report into an existing UIR, and shall create a linked possible-duplicate UIR if any condition fails.

7. The system shall detect field-level contradictions between merged reports and apply three-factor weighted resolution (source reliability, corroboration, recency), displaying unresolved contradictions as ranges and setting the operator_review_required flag.

8. The system shall maintain a Unified Incident Record for each active incident containing: incident_type, location (GPS + display name + source strings), people_involved, urgency, confidence score, source list, chronological timeline, flags, linked UIR IDs, status, and operator action log.

9. The system shall push updated UIRs to all connected dashboard clients via WebSocket within 1.5 seconds of UIR update completion.

10. The dashboard shall display all active UIRs as urgency-coded pins on a Leaflet.js OpenStreetMap of Sri Lanka, updated within 5 seconds of any UIR change.

11. The dashboard shall provide a sortable active incidents list with Critical incidents permanently pinned at the top, and an incident detail panel showing the full UIR.

12. The dashboard shall provide Confirm, Split, Merge, Correct, and Dismiss operator actions, with all actions logged to the UIR audit trail.

13. The dashboard shall display a system status bar showing reports per minute, active UIR count, flagged UIR count, and pipeline latency with an alert if latency exceeds 10 seconds.

14. The system shall log all operator actions with action type, operator ID, timestamp, original field value, and free-text note for post-disaster DMC audit.

15. The system shall output post-event incident summary reports suitable for DMC post-disaster review and resource allocation analysis.

5.3 Non-Functional Requirements

1. Performance: Full pipeline latency from JSON arrival to dashboard render shall not exceed 8 seconds under normal disaster load (up to 10 reports/second). UIR update latency shall not exceed 5 seconds.
2. Reliability: The system shall maintain uptime above 99.5% during active disaster events. Redis queue shall prevent data loss during processing delays.
3. Scalability: The pipeline shall process up to 50 reports per minute without latency degradation. LaBSE encoding shall be batchable and GPU-accelerable for higher loads.
4. Usability: The dashboard shall be usable without prior AI training by DMC operators. Critical incidents shall be identifiable within 3 seconds of opening the dashboard. System Usability Scale (SUS) score shall exceed 70.
5. Security: All report data is processed and stored within a DMC-controlled server environment. No personal caller information is transmitted to external services. LaBSE inference runs locally — no data leaves the system boundary.
6. Maintainability: The location gazetteer shall be updatable via a simple JSON edit without code changes. Epsilon values shall be configurable via environment variables. All pipeline stages shall be independently unit-testable.
7. Accuracy: The system shall achieve clustering precision >90%, recall >85%, Critical false-merge rate <2%, and conflict detection accuracy >80% on the evaluation dataset.

6. COMMERCIALIZATION PLAN

The primary customer is the Disaster Management Centre (DMC), a government agency in Sri Lanka that has a statutory obligation to coordinate disaster response. The primary decision-makers are the Director General of the DMC and the IT and Operations divisions. The system meets a business need of the DMC to improve response time and information overload in disaster response. There is a secondary market in provincial emergency operations centers (EOCs) in 9 provinces in Sri Lanka. There is also an export market in South Asian disaster response agencies, including the Bangladesh National Disaster Management Authority (NDMA), and India's National Disaster Response Force (NDRF), which also deal with multilingual emergency communications in low-resource language environments.

The product has three value propositions that can be measured. These are: Speed – reducing duplicate report processing time from 15-20 minutes to less than 8 seconds per event. Safety – ensuring that critical incidents are not overlooked due to information overload through the urgency-sensitive safety architecture. Intelligence – providing structured incident intelligence to DMC operators instead of raw call records, allowing them to make decisions to deploy resources in seconds.

The revenue streams are in line with the Government Technology (GovTech) SaaS Licensing Model, which is appropriate for the Sri Lanka public sector procurement environment. The revenue streams are split into three tiers: (1) Year 1-2: Research Pilot with DMC under a sponsored research agreement (no license fee; funded by NSF Sri Lanka); (2) Year 3-4: national deployment license to DMC at LKR 3.5M per year (1 year of hosting, support, and updates included); and (3) Year 5+: regional export license to Bangladesh NDMA and India NDRF at USD 50,000–120,000 per year per country. The data insights subscription service provides an additional revenue stream of LKR 1.2M annually at the national level.

The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support. In the national deployment phase (Years 3 to 4), the system is planned to be offered as an annual Software as a Service (SaaS) license to DMC and Provincial EOCs in Sri Lanka, priced annually at an estimated LKR 3.5 million, which also includes system maintenance and infrastructure services. In the regional expansion phase (Year 5 and onward), the system is planned to be adapted and offered to international disaster management organizations such as the Bangladesh National Disaster Management Authority and India's National Disaster Response Force, priced annually between $50,000 and $120,000 as a regional license depending on the scale of deployment and customization required. In addition to the above services offered to DMC and other research partners, an Analytics Add-on service is planned to be offered to DMC and research partners for post-disaster data analysis and insights reporting, priced annually at approximately LKR 1.2 million.

The competitive advantages of the system are: (1) the first-mover advantage in the multilingual emergency fusion marketplace of South Asia—no competing product speaks to Sinhala or Tamil; (2) the Sri Lanka emergency location gazetteer is a proprietary data asset that cannot be replicated without years of linguistics fieldwork; (3) the use of open-source technology stack of LaBSE, DBSCAN, PostgreSQL, and React minimizes the cost of licenses; (4) the urgency-sensitive safety architecture is a patentable new design that has yet to be replicated by competitors.

Three IP assets to be developed: (1) Sri Lanka emergency locations gazetteer to be filed as a data publication and issued a suitable Creative Commons license for research use; (2) urgency-sensitive

DBSCAN clustering algorithm to be filed as a provisional patent with the National Intellectual Property Office of Sri Lanka (NIPO) after system completion; and (3) research paper to be published to claim scientific priority. The university's IP policy and sponsored research agreement between DMC and SLIIT will be applicable.

7. BUDGET AND JUSTIFICATION

Category	Total (LKR)
Software Licenses	60,000
Cloud Services	30,000
Commercialization & IP	20,000
Miscellaneous	10,000
TOTAL 1-YEAR PROJECT BUDGET	**LKR 120000**

Table 4:Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase / Timeline	Component	Responsible Member	Key Tasks	Workload Contribution
Month 1 – Month 2	Project Planning	All Members	Literature Review, Requirement Analysis, System Architecture Design, Dataset Planning	Shared
Month 3 – Month 5	Voice-Based Language Identification, Incident Understanding & Urgency Detection	Member 1	Audio Preprocessing, Speech-to-Text Transcription, Multilingual Language Detection, Panic/Prosody Analysis, Incident Classification, Location Information, Urgency Detection, Generating Structured JSON Output	Core responsibility of Member 1
Month 3 – Month 5	Multilingual Emergency AI Clarification Agent	Member 2	AI Conversation Agent Design, Information Not Provided, Question Generation, Multilingual Conversation Support (Sinhala, Tamil, English), Panic-Aware Conversation, Handling Interruptions and Call Drops, Generating Structured Conversation Summaries	Core responsibility of Member 2
Month 3 – Month 5	Low-Bandwidth Text/SMS	Member 3	Language Detection for Short	Core responsibility of Member 3

	Emergency Intelligence		Messages, Normalization of Short Message Text, Handling Emojis and Abbreviations, Handling Romanized Sinhala and Tamil, Incident Information, Urgency, Generating Structured SMS Intelligence Output	
Month 6 – Month 7	Multimodal Incident Fusion, Timeline Reconstruction & Dashboard	Member 4	Integration of output from components 1-3, incident clustering, duplicate report detection, conflict resolution, creation of a unified incident record, timeline reconstruction, confidence scoring, visualization using a dashboard	Core responsibility of Member 4
Month 8	System Integration	All Members	Integration of all four components, communication via API, testing data flow between modules	Shared
Month 9	System Testing & Evaluation	All Members	Model evaluation, testing, measuring, validation	Shared
Month 10	Dashboard Improvement	Member 4+ Team	Improvement in visualizations, usability testing, refinement	Shared with lead Member 4

Month 11	System Optimization	All Members	Optimization, bug fixes, testing	Shared
Month 12	Documentation & Presentation	All Members	Writing final report, documentation, research paper, final presentation	Shared

Table 5:Work Breakdown Structure

9. GANTT CHART

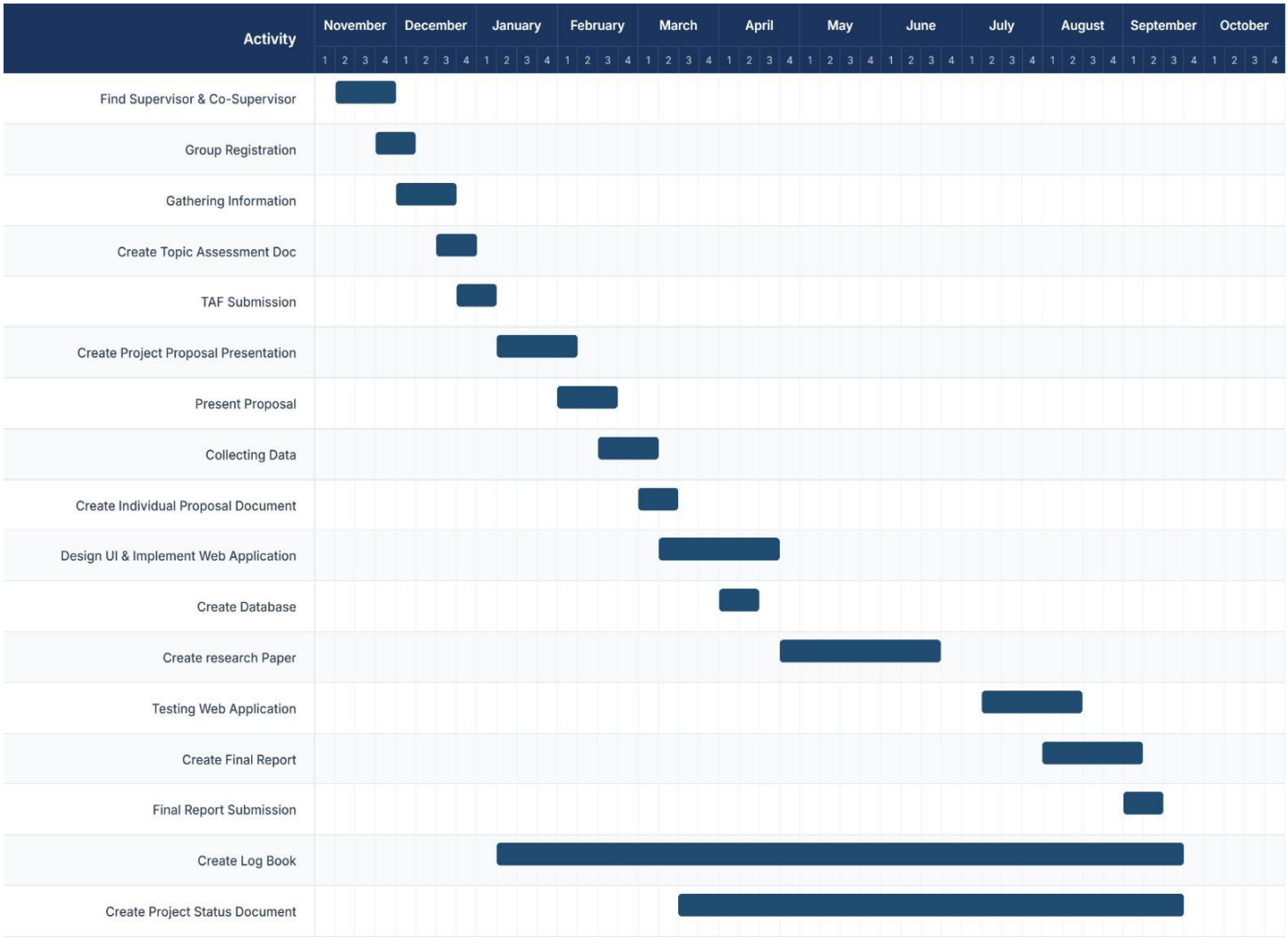


Figure 2: Gantt Chart

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11. APPENDICES

Appendix A: Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
C4 blocked by C2/C3 output delays	**High**	**Medium**	Build and test C4 clustering with synthetic JSON data from Month 3 — fully decoupled from C2/C3
LaBSE insufficient accuracy on SL languages	**Low**	**High**	LaBSE explicitly supports Sinhala and Tamil; evaluate early Month 4; fallback to mBERT if needed
DBSCAN epsilon tuning exceeds time budget	**Medium**	**Medium**	Automate grid search over epsilon combinations on evaluation dataset; begin Month 5
Pipeline latency exceeds 8-second target	**Medium**	**Medium**	Profile each stage individually; pre-cache LaBSE embeddings; use async FastAPI workers
DMC demonstration access delayed	**Medium**	**Low**	Conduct independent evaluation using research team as proxy operators; DMC demo as bonus

Table 6 :Risk analysis

Appendix B: UI Mockup

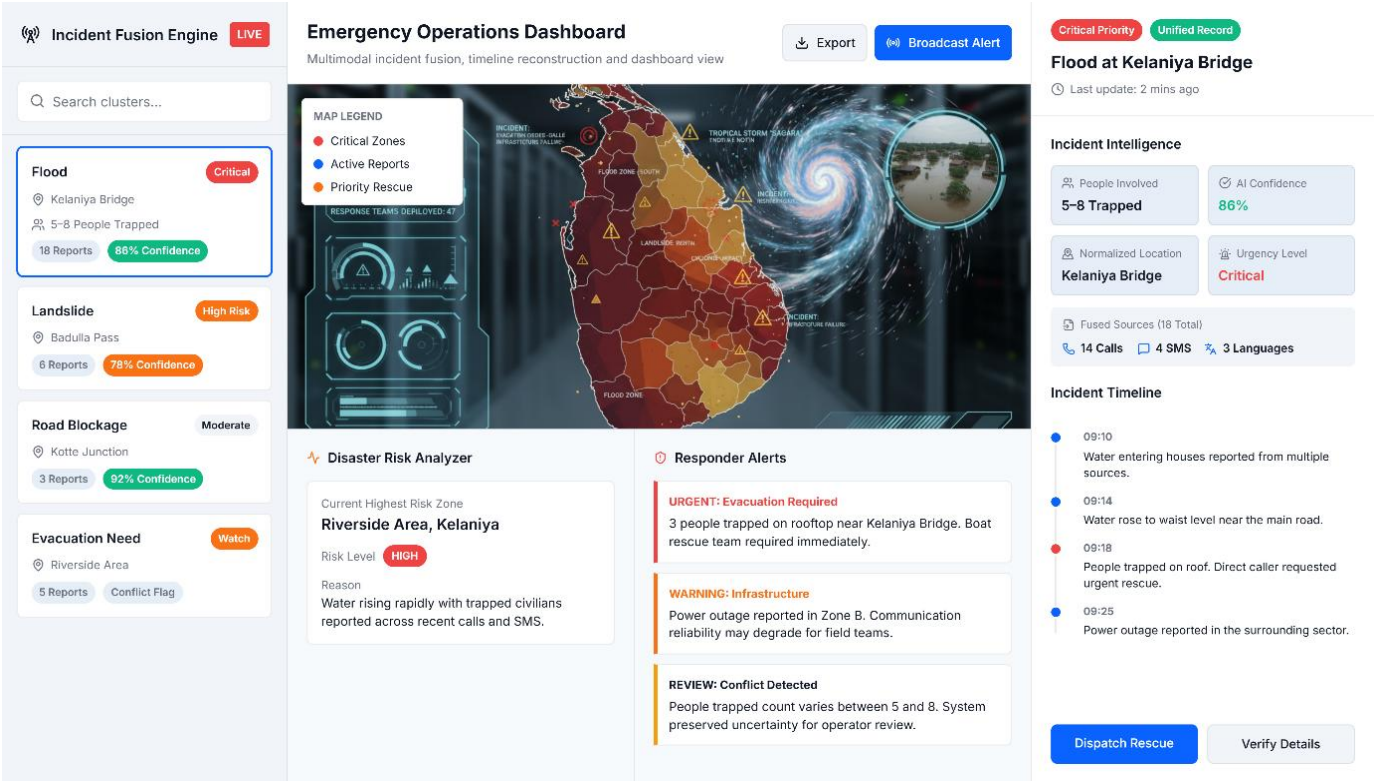


Figure 3:UI Mockup